



Produced-Water Lithium Extraction

TBAC/DA DES + TOPO from Smackover feed basis to staged extraction and screening TEA

Chemical engineering technical review

Smackover Li/Na feed basis → pretreatment → TBAC/DA DES + TOPO → ALAMO surrogate → PrOMMiS/IDAES

May 15, 2026

2026-05-15

Produced-Water Lithium Extraction



Produced-Water Lithium Extraction

TBAC/DA DES + TOPO from Smackover feed basis to staged extraction and screening TEA

Chemical engineering technical review

Smackover Li/Na feed basis → pretreatment → TBAC/DA DES + TOPO → ALAMO surrogate → PrOMMiS/IDAES

May 15, 2026

Open by naming the purpose: this is an internal technical package for connecting Smackover produced-water chemistry to staged extraction and screening economics. Emphasize that the talk is about traceable integration: feed basis, pretreatment boundary, TBAC/DA DES + TOPO, ALAMO, PrOMMiS/IDAES, and TEA.

A pretreated Smackover Li/Na stream connects realistic produced-water chemistry to staged extraction and screening-level economics.

2026-05-15

Produced-Water Lithium Extraction

A pretreated Smackover Li/Na stream connects realistic produced-water chemistry to staged extraction and screening-level economics.

Use this as the one-sentence thesis. The audience should hear that the case study starts from a realistic produced-water feed, removes non-extraction burdens at the pretreatment boundary, and carries only Li/Na transfer variables into the process and cost layers.

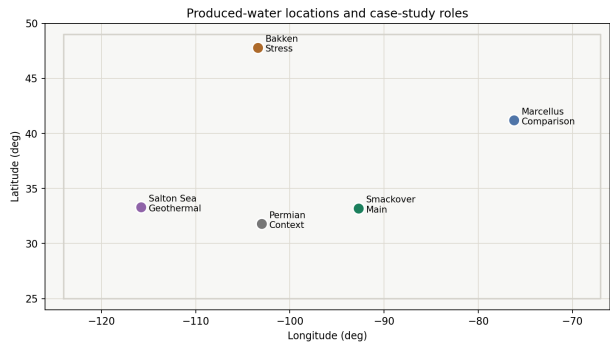
CASE BASIS

2026-05-15

Produced-Water Lithium Extraction
└ Case Basis

CASE BASIS

Smackover Is The Main Produced-Water Case



Case basis

Southern Arkansas Smackover combines meaningful lithium grade with high-salinity major-ion chemistry.

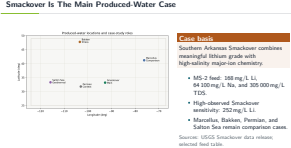
- MS-2 feed: 168 mg/L Li, 64 100 mg/L Na, and 305 000 mg/L TDS.
- High-observed Smackover sensitivity: 252 mg/L Li.
- Marcellus, Bakken, Permian, and Salton Sea remain comparison cases.

Sources: USGS Smackover data release; selected feed table.

2026-05-15

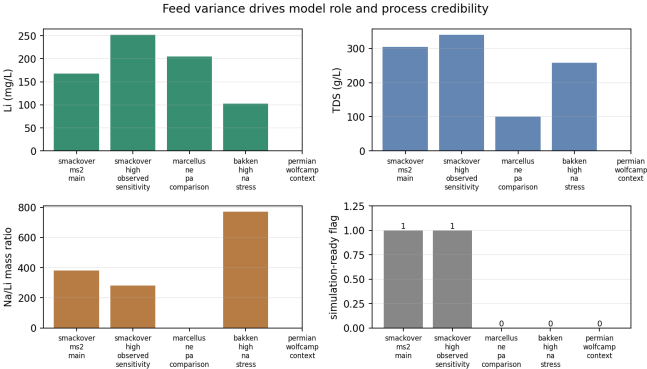
Produced-Water Lithium Extraction

- └ Case Basis
- └ Smackover Is The Main Produced-Water Case



The point is not that Smackover is the only possible brine. It is the strongest primary case because it combines meaningful lithium with major-ion severity. The MS-2 row gives the main feed basis, and the high-Li Smackover row is kept as a clean sensitivity case rather than a new flowsheet.

Feed Variance Sets The Model Role



- Smackover MS-2 anchors the main flowsheet and TEA case.
- The high-Li Smackover row tests feed-grade sensitivity.
- Incomplete major-ion rows stay outside the main process calculation.

Feed	Role
MS-2	main case
High-Li Smackover	sensitivity
Marcellus/Bakken	context

2026-05-15

Produced-Water Lithium Extraction

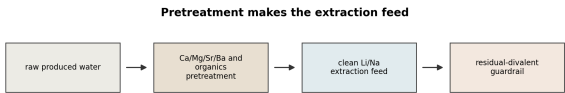
- └ Case Basis

- └ Feed Variance Sets The Model Role

Explain why the model role is narrow. MS-2 is the case, high-Li Smackover is the feed-grade sensitivity, and the other basins provide context. This keeps the presentation from becoming a general produced-water survey while still showing why feed variance matters.



Pretreatment Creates The Li/Na Extraction Feed



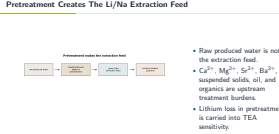
- Raw produced water is not the extraction feed.
- Ca^{2+} , Mg^{2+} , Sr^{2+} , Ba^{2+} , suspended solids, oil, and organics are upstream treatment burdens.
- Lithium loss in pretreatment is carried into TEA sensitivity.

2026-05-15

Produced-Water Lithium Extraction

- └ Case Basis

- └ Pretreatment Creates The Li/Na Extraction Feed



Say that raw produced water is not sent directly to the DES contactor. The extraction model receives a Li/Na feed after divalent removal and oil/solids handling. Pretreatment loss is still represented downstream in the TEA so the boundary is not free.

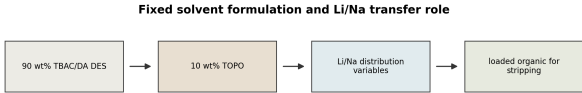
SOLVENT AND TRANSFER SURFACE

2026-05-15

Produced-Water Lithium Extraction
└ Solvent And Transfer Surface

SOLVENT AND TRANSFER SURFACE

TBAC/DA DES + TOPO Is The Active Solvent



- Organic phase: 90 wt% TBAC/DA DES and 10 wt% TOPO.
- TBAC:DA molar ratio: 1:2.
- Temperature: 23 °C.
- Aqueous pH: 10.4.

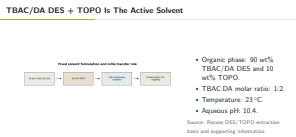
Source: Rezaee DES/TOPO extraction basis and supporting information.

2026-05-15

Produced-Water Lithium Extraction
└ Solvent And Transfer Surface

└ TBAC/DA DES + TOPO Is The Active Solvent

This slide locks the chemistry. The active solvent is TBAC/DA DES with 10 wt% TOPO at the reported pH and temperature. Keep the message simple: the bridge is based on the Rezaee Li/Na extraction response for this solvent system, not a generic extractant screen.



Stage-Response Surface Spans Source And Smackover Domains

Domain	Rows	Use
source LHS	625	extraction anchors
Smackover LHS	625	feed sensitivity
reported anchors	3	Li/Na targets
nominal/corner rows	18	stress bounds

Retained source anchors	
optimized Li extraction	48.57%
optimized Li/Na selectivity	4.41
model-brine Li extraction	51.63%
model-brine Na extraction	9.97%

- Reported extraction responses define the transfer anchors.
- Feed chemistry selects the Smackover case and validity bounds.
- The process calculation reports this paper-anchor transfer basis.

2026-05-15

Produced-Water Lithium Extraction
└ Solvent And Transfer Surface

└ Stage-Response Surface Spans Source And Smackover Domains

Stage-Response Surface Spans Source And Smackover Domains			
Domain	Rows	Use	
source LHS	625	extraction anchors	
Smackover LHS	625	feed sensitivity	
reported anchors	3	Li/Na targets	
nominal/corner rows	18	stress bounds	

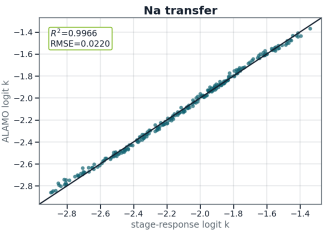
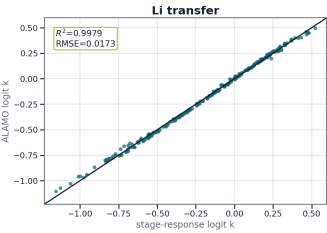
Retained source anchors

optimized Li extraction	48.57%
optimized Li/Na selectivity	4.41
model-brine Li extraction	51.63%
model-brine Na extraction	9.97%

- Reported extraction responses define the transfer anchors.
- Feed chemistry selects the Smackover case and validity bounds.
- The process calculation reports this paper-anchor transfer basis.

Walk through the table as the data contract. The source LHS and Smackover LHS span the response surface, and the reported anchors keep the model tied to the Rezaee extraction values. The important phrase is paper-response bridge: it is compact, traceable, and process-facing.

IDAES ALAMO Surrogate Preserves Li/Na Transfer Trends



- Inputs: Li concentration, Na concentration, O/A ratio, and TOPO loading.
- Outputs: logit-transformed Li and Na transfer coefficients.
- $R^2 = 0.9979$ for Li and $R^2 = 0.9966$ for Na on the held-out table.

Output	RMSE	max error
logit k_{Li}	0.0173	0.0555
logit k_{Na}	0.0220	0.0779

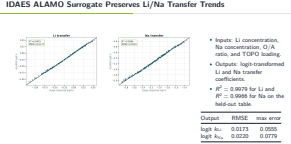
2026-05-15

Produced-Water Lithium Extraction

└ Solvent And Transfer Surface

└ IDAES ALAMO Surrogate Preserves Li/Na Transfer Trends

The ALAMO result is here to justify the compact transfer surface. The held-out parity is strong for both Li and Na in logit transfer space, so the surrogate is appropriate for the staged extraction calculations. Do not overstate this as a direct thermodynamic proof.



Paper-Response Variables Feed The Process Model

$$D_i = \frac{E_i}{100 - E_i} \frac{1}{O/A}$$
$$S_{Li/Na} = \frac{D_{Li}}{D_{Na}}$$



Rezaee anchors, $O/A = 1$

Optimized 10 wt% TOPO: $E_{Li} = 48.57\%$, $S_{Li/Na} = 4.41$, so $E_{Na} = 11.01\%$.
Model-brine row: $E_{Li} = 51.63\%$ and $E_{Na} = 9.97\%$.

2026-05-15

Produced-Water Lithium Extraction
└ Solvent And Transfer Surface

└ Paper-Response Variables Feed The Process Model

Paper-Response Variables Feed The Process Model

$$D_i = \frac{E_i}{100 - E_i} \frac{1}{O/A}$$
$$S_{Li/Na} = \frac{D_{Li}}{D_{Na}}$$

Rezaee anchors, $O/A = 1$

Optimized 10 wt% TOPO: $E_{Li} = 48.57\%$, $S_{Li/Na} = 4.41$, so $E_{Na} = 11.01\%$.
Model-brine row: $E_{Li} = 51.63\%$ and $E_{Na} = 9.97\%$.

```
graph LR; A[Smackover Li/Na feed] --> B[Divalent pretreatment]; B --> C[TBAC/DA DES + TOPO]; C --> D[ALAMO transfer surface]; D --> E[PrOMMiS/IDAES stages];
```

This is the equation slide for the process bridge. Extraction percent is converted to distribution coefficients, and Li/Na selectivity is the ratio of those coefficients. The Rezaee anchors give the numerical grounding; the flow diagram shows where the variables enter PrOMMiS/IDAES.

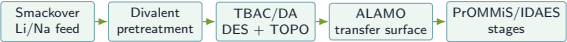
PROCESS AND ECONOMICS

2026-05-15

Produced-Water Lithium Extraction
└ Process And Economics

PROCESS AND ECONOMICS

PrOMMiS/IDAES Uses Active Li And Na Transfer Terms



Rezaee extraction responses are converted to Li/Na transfer variables for process calculations

Stages	Li recovery	Na co-transfer
1	45.681%	10.714%
3	83.770%	29.008%
5	95.100%	43.740%

- The process layer uses PrOMMiS SolventExtraction objects with IDAES MSContactor units.
- Lithium and sodium material-transfer terms are active.
- Chloride transfer is held at zero inside the Li/Na extraction boundary.

2026-05-15

Produced-Water Lithium Extraction

- └ Process And Economics

└ PrOMMiS/IDAES Uses Active Li And Na Transfer Terms

Smackover Li/Na feed

Divalent pretreatment

TBAC/DA DES + TOPO

ALAMO transfer surface

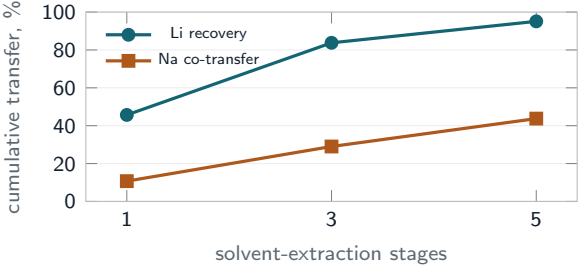
PrOMMiS/IDAES stages

Stage	Li recovery	Na co-transfer
1	45.681%	10.714%
3	83.770%	29.008%
5	95.100%	43.740%

- The process layer uses PrOMMiS SolventExtraction objects with IDAES MSContactor units.
- Lithium and sodium material-transfer terms are active.
- Chloride transfer is held at zero inside the Li/Na extraction boundary.

Emphasize the process-model handoff: Li and Na transfer are active terms, while chloride is held outside the extraction boundary. The staged results are not a standalone chemistry claim; they are what happens when the paper-response transfer variables are carried into the PrOMMiS/IDAES stage structure.

Three Stages Recover 83.77% Lithium



- Smackover MS-2 at $O/A = 1$ and 10 wt% TOPO reaches 83.77% cumulative lithium recovery after three stages.
- Sodium co-transfer is 29.01% at the same condition.
- Five stages raise lithium recovery to 95.10% with a larger sodium burden.

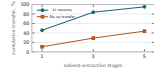
2026-05-15

Produced-Water Lithium Extraction

└ Process And Economics

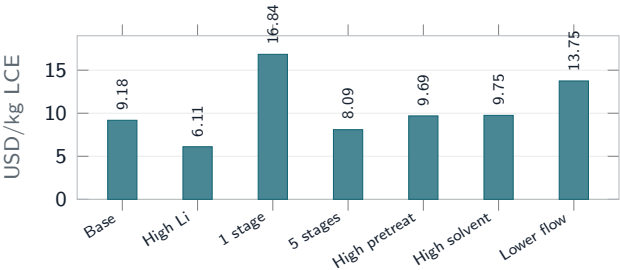
└ Three Stages Recover 83.77% Lithium

This is the recovery slide. One stage gives the first extraction step, three stages is the main design point, and five stages shows the recovery ceiling with higher sodium co-transfer. The 83.77% number is the main process result to carry forward into the TEA.



- Smackover MS-2 at $O/A = 1$ and 10 wt% TOPO reaches 83.77% cumulative lithium recovery after three stages.
- Sodium co-transfer is 29.01% at the same condition.
- Five stages raise lithium recovery to 95.10% with a larger sodium burden.

Screening TEA Centers At 9.18 USD/kg LCE



- Base Smackover intensity is 9.18 USD/kg lithium carbonate equivalent.
- High-Li feed sensitivity lowers the value to 6.11 USD/kg.
- One stage raises the value to 16.84 USD/kg through lower recovery.

2026-05-15

Produced-Water Lithium Extraction └ Process And Economics

└ Screening TEA Centers At 9.18 USD/kg LCE



Present the TEA as screening-level prioritization. The base case is 9.18 USD/kg LCE, high feed grade moves the number down, and one stage moves it up by losing recovery. The value of the slide is ranking the levers, not claiming plant cost certainty.

Economic Levers Are Feed Grade, Stages, And Losses

Scenario	Recovery	Cost	p10–p90
base Smackover	79.58%	9.18	8.81–12.43
high-Li Smackover	79.72%	6.11	5.92–8.26
one stage	43.40%	16.84	16.15–22.52
five stages	90.35%	8.09	7.77–10.74
high pretreatment loss	75.39%	9.69	8.93–12.57
high solvent loss	79.58%	9.75	9.00–12.59
lower throughput	79.58%	13.75	12.65–18.96

Cost values in USD/kg lithium carbonate equivalent.

Screening boundary

The TEA compares operating sensitivities for prioritization; it does not replace equipment design, residence-time sizing, solvent-loss experiments, or product-pricing analysis.

- Feed lithium grade is the strongest favorable lever.
- Stage count trades lithium recovery against sodium loading.
- Fixed annual charges dominate the lower-throughput case.

2026-05-15

Produced-Water Lithium Extraction

Process And Economics

Economic Levers Are Feed Grade, Stages, And Losses

Scenario	Recovery	Cost	p10–p90
base Smackover	79.58%	9.18	8.81–12.43
high-Li Smackover	79.72%	6.11	5.92–8.26
one stage	43.40%	16.84	16.15–22.52
five stages	90.35%	8.09	7.77–10.74
high pretreatment loss	75.39%	9.69	8.93–12.57
high solvent loss	79.58%	9.75	9.00–12.59
lower throughput	79.58%	13.75	12.65–18.96

Screening boundary

The TEA compares operating sensitivities for prioritization; it does not replace equipment design, residence-time sizing, solvent-loss experiments, or product-pricing analysis.

- Feed lithium grade is the strongest favorable lever.
- Stage count trades lithium recovery against sodium loading.
- Fixed annual charges dominate the lower-throughput case.

Use this slide to explain tradeoffs. Better lithium grade and enough stages improve cost intensity; pretreatment loss, solvent loss, and lower throughput move the result in the wrong direction. The p10–p90 band is useful for showing that feed and stage choices dominate the screening sensitivity.

THERMODYNAMIC EXTENSION

2026-05-15

Produced-Water Lithium Extraction
└─ Thermodynamic Extension

THERMODYNAMIC EXTENSION

Case-Study Coverage Matrix

Lane	Current basis	Process-facing use
feed basis	Smackover MS-2 and high-Li Smackover sensitivity	defines the main and feed-grade sensitivity cases
paper-response bridge	Rezaee Li/Na extraction responses with <i>O/A</i> conversion	supplies D_{Li} , D_{Na} , and $S_{Li/Na}$ for staged extraction
ALAMO surrogate	held-out parity for logit k_{Li} and logit k_{Na}	provides the compact transfer surface used by the process model
PrOMMiS/IDAES stages	PrOMMiS/IDAES staged-extraction terms for Li and Na across 1, 3, and 5 stages	converts transfer variables to outlet recovery and raffinate behavior
screening TEA/UQ	feed grade, stage count, pretreatment loss, solvent loss, and throughput	ranks operating levers for sprint approval
direct ePC-SAFT replacement path	accepted Li/Na transfer rows	replaces the paper-response bridge without changing process inputs

Process-facing rows remain stable as the thermodynamic basis is exchanged.

2026-05-15

Produced-Water Lithium Extraction └ Thermodynamic Extension

└ Case-Study Coverage Matrix

Lane	Current basis	Process-facing use
feed basis	Smackover MS-2 and high-Li Smackover sensitivity	defines the main and feed-grade sensitivity cases
paper-response bridge	Rezaee Li/Na extraction responses with <i>O/A</i> conversion	supplies D_{Li} , D_{Na} , and $S_{Li/Na}$ for staged extraction
ALAMO surrogate	held-out parity for logit k_{Li} and logit k_{Na}	provides the compact transfer surface used by the process model
PrOMMiS/IDAES stages	PrOMMiS/IDAES staged-extraction terms for Li and Na across 1, 3, and 5 stages	converts transfer variables to outlet recovery and raffinate behavior
screening TEA/UQ	feed grade, stage count, pretreatment loss, solvent loss, and throughput	ranks operating levers for sprint approval
direct ePC-SAFT replacement path	accepted Li/Na transfer rows	replaces the paper-response bridge without changing process inputs
Process-facing rows remain stable as the thermodynamic basis is exchanged.		

This is the scope-control slide. The current package covers the feed basis, paper-response bridge, ALAMO surrogate, PrOMMiS/IDAES stages, and screening TEA/UQ. For interim ePC-SAFT support, cite the Rezaee 2026 26-row reactive-equilibrium replay: the package evaluates the aqueous ePC-SAFT and organic PC-SAFT activity terms, and the calibrated actual-row replay has median $\ln Q - \ln K$ offsets of -0.23 for Li and 0.20 for Na. That supports the replacement path, but it is not a direct Li/Na transfer-row result.

The case study is a screening-quality bridge from Smackover feed chemistry to PrOMMiS/IDAES staged extraction and TEA, ready to anchor an internal ePC-SAFT-to-PrOMMiS/IDAES integration sprint.

2026-05-15

Produced-Water Lithium Extraction
└─ Thermodynamic Extension

The case study is a screening-quality bridge from Smackover feed chemistry to PrOMMiS/IDAES staged extraction and TEA, ready to anchor an internal ePC-SAFT-to-PrOMMiS/IDAES integration sprint.

Say the ask plainly: approve a focused internal sprint that connects accepted ePC-SAFT Li/Na transfer rows to the existing PrOMMiS/IDAES interface. The deck is already organized so the paper-response bridge can be exchanged without changing the process inputs.

- 1. Smackover MS-2 is the main produced-water feed; the high-Li Smackover row defines the feed-grade sensitivity.
- 2. TBAC/DA DES + 10 wt% TOPO maps lithium and sodium transfer through ALAMO and into staged solvent extraction.
- 3. Three stages recover 83.77% lithium before the TEA pretreatment-loss adjustment.
- 4. The base screening TEA is 9.18 USD/kg LCE, with feed grade and stage count controlling the largest movement.
- 5. Ask: approve an internal ePC-SAFT-to-PrOMMiS/IDAES integration and validation sprint.

2026-05-15

Produced-Water Lithium Extraction

└ Thermodynamic Extension

└ Technical Message

Read this slide as the closing script. The first four bullets summarize the technical chain and the fifth bullet is the decision request. If asked about current ePC-SAFT data, use the results folder carefully: it supports Rezaee-style activity-term replay and calibrated actual rows, while direct transfer rows remain the next accepted input to the process bridge.

- Technical Message
- 1. Smackover MS-2 is the main produced-water feed; the high-Li Smackover row defines the feed-grade sensitivity.
 - 2. TBAC/DA DES + 10 wt% TOPO maps lithium and sodium transfer through ALAMO and into staged solvent extraction.
 - 3. Three stages recover 83.77% lithium before the TEA pretreatment-loss adjustment.
 - 4. The base screening TEA is 9.18 USD/kg LCE, with feed grade and stage count controlling the largest movement.
 - 5. Ask: approve an internal ePC-SAFT-to-PrOMMiS/IDAES integration and validation sprint.

Supplemental Extension Context

Family	Why it belongs near produced water	Needed before modeling
CMM/REE ions	oilfield brines can carry critical-mineral value beyond Li	measured composition, aqueous speciation, and ion parameters
divalent ions	Ca, Mg, and Sr set pretreatment load and leakage risk	pretreatment data and extraction leakage bounds
hydrocarbons	oil-bearing water adds phase behavior and solvent-contact risks	oil-phase assay and phase-equilibrium data
ePC-SAFT extension	electrolyte and Born/dielectric terms give a thermodynamic route	accepted parameters and phase data for each species family

Boundary

Li/Na extraction remains the present case. CMM, REE, and hydrocarbon rows define where additional data would expand the same structure.

Use

The extension list guides what to measure before adding species, reactions, or oil-phase terms to the thermodynamic model.

USGS Energy Waste Science; critical-mineral produced-water review; electrolyte PC-SAFT permittivity and ePC-SAFT Born/dielectric literature.

2026-05-15

Produced-Water Lithium Extraction

Thermodynamic Extension

Supplemental Extension Context

Keep this slide supplemental. It says CMM, REE, divalent ions, and hydrocarbons are natural produced-water extensions, but each needs measured composition, phase data, parameters, and extraction chemistry before joining the case study. Do not imply these systems are already solved by the Li/Na model.

Supplemental Extension Context		
Family	Why it belongs near produced water	Needed before modeling
CMM/REE ions	oilfield brines can carry critical-mineral value beyond Li	measured composition, aqueous speciation, and ion parameters
divalent ions	Ca, Mg, and Sr set pretreatment load and leakage risk	pretreatment data and extraction leakage bounds
hydrocarbons	oil-bearing water adds phase behavior and solvent-contact risks	oil-phase assay and phase-equilibrium data
ePC-SAFT extension	electrolyte and Born/dielectric terms give a thermodynamic route	accepted parameters and phase data for each species family
Boundary		Use
Li/Na extraction remains the present case. CMM, REE, and hydrocarbon rows define where additional data would expand the same structure.		The extension list guides what to measure before adding species, reactions, or oil-phase terms to the thermodynamic model.
USGS Energy Waste Science; critical-mineral produced-water review; electrolyte PC-SAFT permittivity and ePC-SAFT Born/dielectric literature.		

- Rezaee et al. 2025: TBAC/DA DES + TOPO extraction response basis.
- Rezaee et al. 2026: PC-SAFT/ePC-SAFT parameter and reaction-equilibrium support.
- U.S. Geological Survey southern Arkansas Smackover brine data release, DOI: 10.5066/P9QPRYZN.
- USGS Energy Waste Science and produced-water lithium/mineral recovery context.
- Critical-mineral source potential review for U.S. oil and gas produced waters.
- Electrolyte PC-SAFT permittivity and ePC-SAFT Born/dielectric literature for the extension basis.
- Produced-water feed table, ALAMO validation table, PrOMMiS/IDAES staged-extraction table, and screening TEA table from the case-study result tables.

2026-05-15

Produced-Water Lithium Extraction
└ Thermodynamic Extension

└ Sources

Use sources only if the audience asks where the basis comes from. Rezaee supplies extraction and thermodynamic-modeling inputs, USGS supplies the produced-water context, and the case-study result tables supply the bridge, stage, and TEA values used in the deck.

Sources
• Rezaee et al. 2025: TBAC/DA DES + TOPO extraction response basis. • Rezaee et al. 2026: PC-SAFT/ePC-SAFT parameter and reaction-equilibrium support. • U.S. Geological Survey southern Arkansas Smackover brine data release, DOI: 10.5066/P9QPRYZN. • USGS Energy Waste Science and produced-water lithium/mineral recovery context. • Critical-mineral source potential review for U.S. oil and gas produced waters. • Electrolyte PC-SAFT permittivity and ePC-SAFT Born/dielectric literature for the extension basis. • Produced-water feed table, ALAMO validation table, PrOMMiS/IDAES staged-extraction table, and screening TEA table from the case-study result tables.